

Capitolis CCR Monte Carlo Engine

Methods, Results, and Feedback Addressed

Berkeley MFE Industry Project

September 27, 2026

Book: 16 trades, 3 counterparties (A/B/C)
8 equity TRS, 4 bond forwards, 4 bond TRS
37 equities (US + Tokyo) + USDJPY

Engine: full Monte Carlo, real data throughout

- ▶ Equities/FX: correlated GBM, real 3y realised vols and correlations
- ▶ USD rates: one-factor Hull-White, swaption-calibrated mean reversion, realised long-end vol
- ▶ JPY rates: a second Hull-White factor (real JPY OIS curve)
- ▶ Latin Hypercube sampling, common random numbers throughout

Exposure definition (the brief):

$$\max(V(t+10\text{bd}) - V(t-1\text{bd}), 0)$$

signed prior-day VM, no IM/threshold, netted per counterparty

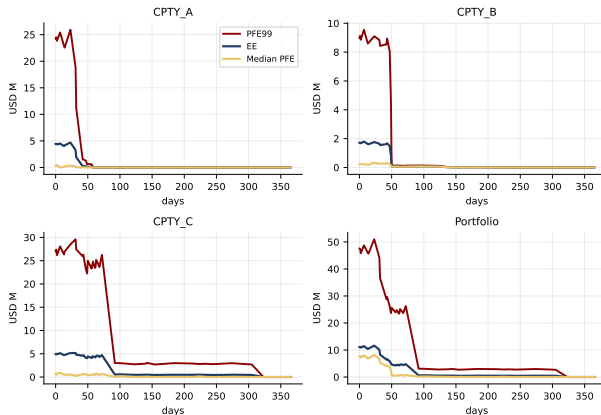
Deliverables: 77-page report, this deck, an Excel parametric-benchmark workbook, 168 automated regression tests

Headline results (5,000 scenarios, close-out exposure)

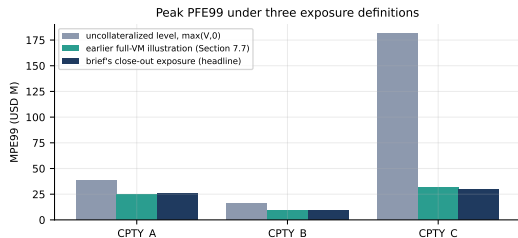
	MPE99	Peak EE
CPTY_A (equity)	\$25.9M	\$4.7M
CPTY_B	\$9.6M	\$1.8M
CPTY_C (rates)	\$29.6M	\$5.2M
Portfolio	\$51.0M	\$11.6M

Portfolio MPE99 is 78% of the sum of counterparties (\$65.1M): no netting across counterparties, but exposures still overlap in time and are positively dependent.

CVA \$11.7k (close-out) / \$247k (level);
SA-CVA capital \$0.10M / \$2.23M; SA-CCR
EAD \$260M.

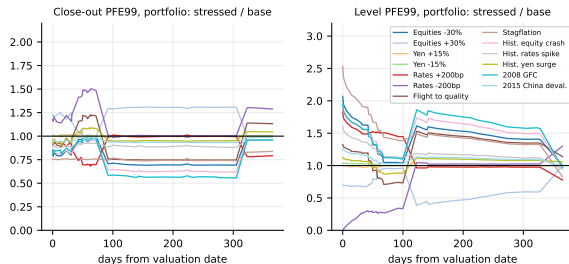


Close-out vs. level exposure



- ▶ **Close-out** (the brief): only the 10-day move from a margined start. Small because margin already covers today's mark-to-market.
- ▶ **Level** (secondary view): the whole mark-to-market at risk, no margin assumed. Portfolio MPE \$201.9M – an order of magnitude larger, driven by CPTY_C's bond forward.
- ▶ The two numbers *bracket* the truth: which one applies depends on whether the trade is actually margined.

Stress testing: structured, not a paragraph



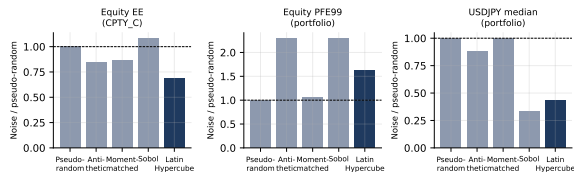
13 scenarios, same random numbers as the base case:

- ▶ 8 round-number shocks (equities $\pm 30\%$, yen $\pm 15\%$, rates $\pm 200\text{bp}$, flight-to-quality, stagflation)
- ▶ 5 historical replays: worst equity/rates/yen windows 2014–2026, **plus 2008 GFC and Aug 2015 China deval.**

2008 finding: close-out MPE99 -13% , level $+56\%$ – inside the range the hypothetical scenarios already covered. Not leaving out the tail.

Latin Hypercube: where it helps, and a genuine extension

Sampling scheme vs Greeks noise: no consistent winner

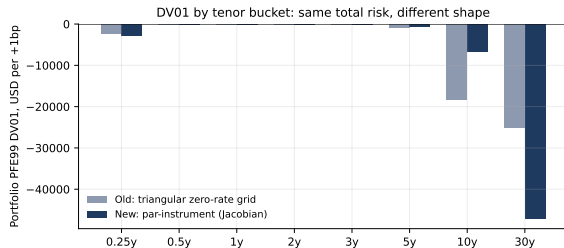


Base measures (Section 5.4): Latin Hypercube is the best of 5 schemes tested.

Greeks noise: mixed. LHS cuts noise on EE deltas (15–85%) but not consistently on tail-quantile (PFE99) deltas.

New: LHS as the sensitivity. In one-factor Hull-White, $r(t) = x(t) + \alpha(t)$: the simulated $x(t)$ is unchanged by a curve bump. So a rate Greek is a closed-form shift of the *same* simulated paths – no resimulation. Validated to 10^{-15} relative precision against a true resimulation.

DV01 via a Jacobian: bump the curve, not just the zero rate

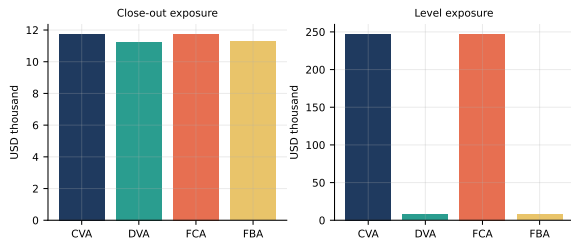


Old method: bump the *fitted* zero curve at 8 hand-picked tenors.

New method: bump the curve's own ~45 native construction pillars (SOFR futures + Bloomberg long end), grouped into the same 8 buckets – the real par-instrument sensitivity.

Finding: the 30y bucket carries 82% of portfolio DV01 (new) vs. 54% (old) – the old grid was smearing the 20–50y risk, where the 2049 bond forward actually lives, into the 10y bucket. Bucket sums match the parallel bump to within 0.1% either way.

CVA, DVA, FVA, and now KVA



Why we build this in detail: the value is the sensitivity chain (CVA01, rating table, SA-CVA weighted sensitivities), not the headline number.

New: KVA. $K(t) = 8\% \times RW \times \alpha \times EPE(t)$, discounted, scaled by an assumed cost of capital (8/10/12% swept, none given).

At 10% CoC: KVA \$22.4k (close-out) / \$470k (level) – larger than CVA (\$11.7k / \$247k) because it scales the whole EPE profile, not a small IG default probability. Explicitly illustrative.

Independent parametric benchmarks (Excel)

A separate, formula-only Excel workbook checks the engine against a hand calculator, using only public data:

- ▶ **Equity/FX vol:**
 $\text{=STDEV}(\text{LN}(P_t/P_{t-1})) * \text{SQRT}(252)$ on raw yfinance closes matches the calibrated model vol to $\sim 0.1\%$.
- ▶ **Rate vol:** same idea on FRED Treasury yields; the closed-form Hull-White formula $\sigma(1 - e^{-aT})/(aT)$ lands within 5–10% of realised at the fitted tenors.
- ▶ **Simulation cross-check:** analytic vol $\times \sqrt{T}$ vs. the standard deviation actually measured across 5,000 simulated scenarios: matches to 1–2%.
- ▶ **Vega bump:** bumping a USD name's vol $\pm 1\%$, the Monte Carlo vega of a simple ATM position matches the closed-form (undiscounted) Black vega to $\sim 7\%$.

`docs/Capitolis_CCR_Parametric_Benchmarks.xlsx` – every number is a live formula, not a pasted value.

Feedback addressed – earlier meeting

Feedback	How it was incorporated
Robust stress testing (data, inputs, sensitivity, across path/products) Sensitivities: method, cost, faster alternative	Done: 13 scenarios, same random numbers, read at every date/trade/counterparty (Section 7.10) Done: bump-and-reprice with CRN; pathwise for equity/FX (1000x faster, validated); adjoint noted as future work (Section 9)
PCA explainability and speed	Done: factor loadings explained; found <i>no</i> speed gain as implemented, reported honestly (Section 4.4)
Final results: A=equity, C=rates, bond-equity relation, portfolio additivity CVA walkthrough and set-up for next session	Done: sensitivity table, rank correlations, and the “no netting across counterparties” explanation (Section 7.2) Done: full walkthrough, 6 open questions listed for next session (Section 8.8)

Feedback addressed – this meeting

Feedback	How it was incorporated
Add 2008, more testing periods	Done: 2008 GFC + Aug 2015 China deval, public data back to 2007 (Section 7.10)
Use LHS to calculate sensitivities	Done: exact analytic rate-bump reusing the same simulated draws, validated to 10^{-15} (Section 9.9)
KVA addition	Done: parametric capital charge, CoC grid (Section 8.6)
CVA is mostly for sensitivities/Greeks	Done: reframed explicitly; sensitivity chain is the deliverable (Section 8.8)
DV01 via Jacobian, bump the curve not just zero rate	Done: par-instrument buckets from the curve's ~ 45 native pillars (Section 9.8)
Excel parametric test: bump equity vol	Done: closed-form Black vega vs. Monte Carlo vega, live formulas (Section 5.6)

- ▶ Which exposure definition should price CVA in production: margined close-out, or level?
- ▶ Real ratings or CDS for CPTY_A/B/C – the BBB proxy currently drives every credit number.
- ▶ Capitolis' own credit and funding spread for DVA/FVA; is DVA recognised at all (Basel ignores it)?
- ▶ A real capital methodology and cost of capital for KVA (currently swept over an assumed 8/10/12% grid).
- ▶ Wrong-way risk: independence of exposure and default is assumed – wanted?
- ▶ Adopt the two-factor rate model (G2++, compared but not default: changes portfolio MPE99 by only -3%)?

Thank you

Questions?